Search for Extraterrestrial Intelligence in the New Era

Bachelor's Thesis Defense Report

Wang Yiyu

School of Physics and Astronomy, Sun Yat-sen University

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Sun Yat-sen University

Candidate: Wang Yiyu | Student ID: 22344140

Supervisor: Prof. Zhang Yong

Contents

1. Introduction & Research Background
2. Methodology & Three-Dimensional Framework
3. Historical Evolution of SETI (1960–2025)
4. SETI vs. Traditional Observational Science
5. SKA's Three-Dimensional Capability Leap
6. Conclusions & Future Perspectives

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1 Introduction & Research Background

2 Methodology & Three-Dimensional Framework

3 Historical Evolution of SETI (1960–2025)

4 SETI vs. Traditional Observational Science

5 SKA's Three-Dimensional Capability Leap

6 Conclusions & Future Perspectives

7 Q & A

SYSU

Research Background

Historical Origins

- 1959: Cocconi & Morrison published the foundational paper in *Nature*, systematically arguing that interstellar communication signals could be observable
- 1960: Frank Drake launched Project Ozma at Green Bank Observatory — the birth of modern SETI
- Evolution from narrow-band listening, Cold War-era grand designs, to contemporary technosignature search frameworks

Core Scientific Difficulties

- Extremely large search space with highly uncertain signal forms
- Chronic observational coverage gaps — null results difficult to constrain
- Involves extremely rare targets, parameter spaces, and frontier methodology

Research Significance & Core Question

Definition 1.1 Central Thesis

How can SETI transform from a controversial exploratory topic into a testable observational science?

Approach

- Treat sky coverage, frequency coverage, and sensitivity as three critical dimensions
- Establish a unified observational coordinate system across historical stages
- Demonstrate why SKA may serve as the **facility inflection point** for new-era SETI
- Push SETI toward observational science characterized by parameter space constraints

1 Introduction & Research Background

2 Methodology & Three-Dimensional Framework

3 Historical Evolution of SETI (1960–2025)

4 SETI vs. Traditional Observational Science

5 SKA's Three-Dimensional Capability Leap

6 Conclusions & Future Perspectives

7 Q & A

SYSU

Data Sources & Literature Foundation

Four Types of Materials

- **ADS database**: 1960–2025 SETI publications for staged evolution analysis
- **Classical literature**: from Cocconi-Morrison to the technosignature discourse
- **Representative observations**: Breakthrough Listen's 1327-star search over 1.10–3.45 GHz
- **SKAO official documents**: system specifications and performance parameters

Dual Bibliographic Datasets

- **Strict core set**: titles/abstracts containing SETI, technosignature keywords → **778 samples** (peer-reviewed astronomy)
- **Comprehensive set**: expanded to astronomy & physics → **1,519 samples** for broader trends

Three-Dimensional Analysis Framework

1. Sky Coverage — Cumulative observed sky area (deg^2)

- Targeted monitoring: approximate by $N_{tgt} \times \Omega_b$ (primary beam area)
- Survey programs: use publicly reported area orders of magnitude
- Emphasis on order-of-magnitude comparison

2. Frequency Coverage — Actual searched frequency range (GHz)

- Distinguish nominal instrument capability from actual program search range
- $\Delta f = f_{max} - f_{min}$: what was *actually* searched, not what the instrument *could* cover

3. Sensitivity — Minimum detectable signal capability

- Strictly uses A_{eff}/T_{sys}, SEFD, and radiometer equation

1 Introduction & Research Background

2 Methodology & Three-Dimensional Framework

3 Historical Evolution of SETI (1960–2025)

4 SETI vs. Traditional Observational Science

5 SKA's Three-Dimensional Capability Leap

6 Conclusions & Future Perspectives

7 Q & A

SYSU

Four Stages of SETI Development

Stage 1 (1960s–70s): Pioneer Era

- Cocconi-Morrison (1959); Drake's Ozma (1960)
- 4 papers total, $\bar{N} \approx 0.36$/yr
- Narrow-band pathway established

Stage 2 (1971–92): Expansion Era

- NASA HRMS program
- 381 papers, $\bar{N} \approx 17.3$/yr
- Growth rate $g = 9425\%$ from institutionalization

Stage 3 (1993–2015): Transition Era

- Post-NASA decentralized growth
- 563 papers, $\bar{N} \approx 24.5$/yr
- Cross-disciplinary accumulation

Stage 4 (2016–25): Renaissance Era

- Breakthrough Listen, FAST participation
- 571 papers, $\bar{N} \approx 57.1$/yr
- 133% annual increase vs. Stage 3

Four-Stage Diagram & Literature Statistics

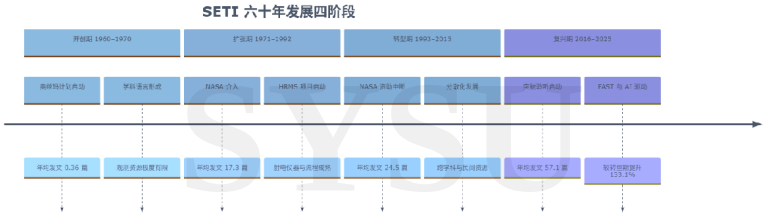
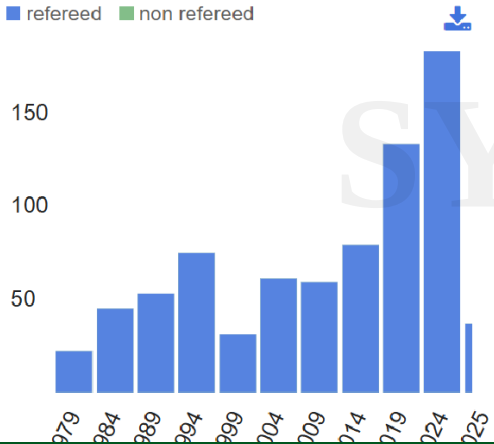
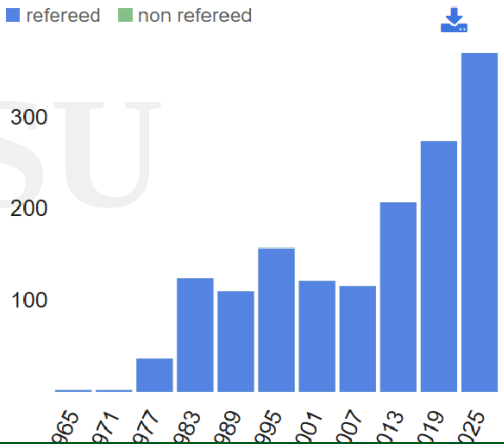


Figure 3.1: Sixty years of SETI development in four stages

Literature Statistics: Strict Core & Comprehensive Sets





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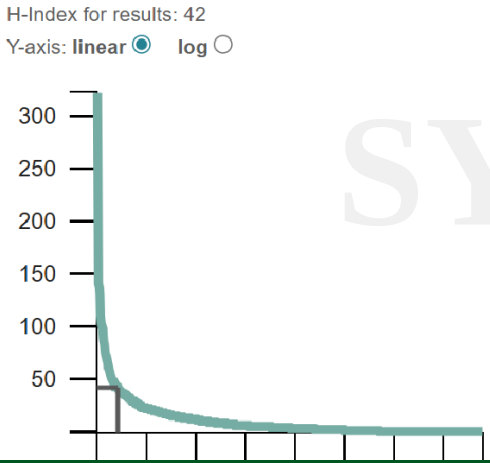
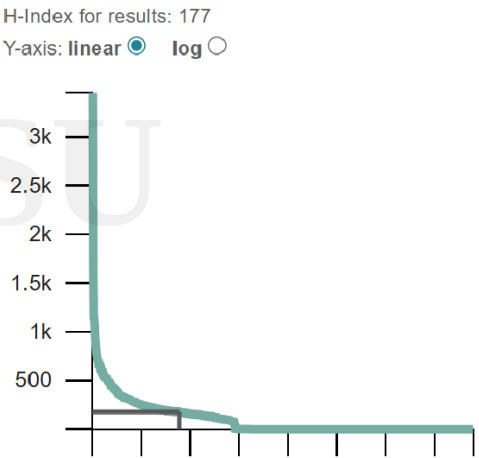
Core Citation & Readership Metrics





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Why Radio Has Long Dominated SETI

Physical Advantages

- Lower photon energy at radio frequencies → efficient for long-range information transmission
- Minimal atmospheric attenuation at radio wavelengths
- Interstellar medium attenuation relatively controlled

Technical Advantages

- Mature pipeline: narrow-band detection, Doppler correction, RFI rejection, repeat observation
- High equipment overlap with conventional radio astronomy — enables "commensal" (shared) observing
- Drake Equation provides a quantitative framework

1 Introduction & Research Background

2 Methodology & Three-Dimensional Framework

3 Historical Evolution of SETI (1960–2025)

4 SETI vs. Traditional Observational Science

5 SKA's Three-Dimensional Capability Leap

6 Conclusions & Future Perspectives

7 Q & A

SYSU

Paradigm Differences: SETI vs. Traditional Astronomy

Dimension	Traditional Astronomy	SETI
Target	Objects known to exist	Existence itself uncertain
Null result	Can directly constrain model parameters	Only excludes local hypotheses
Positive threshold	Repeated observation + model fitting	Must exclude RFI, instrument artifacts, all natural processes
Evaluation	Precision & sample size	Public focuses on "discovery or not"
Funding	Embedded in stable observational systems	Relies on special initiatives

Cosmology as a Reference

Cosmology transformed from "speculative" to "precision science" through observational advances (Hubble, COBE, Planck). SETI needs the same: turning vague null results into

1 Introduction & Research Background

2 Methodology & Three-Dimensional Framework

3 Historical Evolution of SETI (1960–2025)

4 SETI vs. Traditional Observational Science

5 SKA's Three-Dimensional Capability Leap

6 Conclusions & Future Perspectives

7 Q & A

SYSU

Frequency: From Limited Windows to Facility-Level Wideband

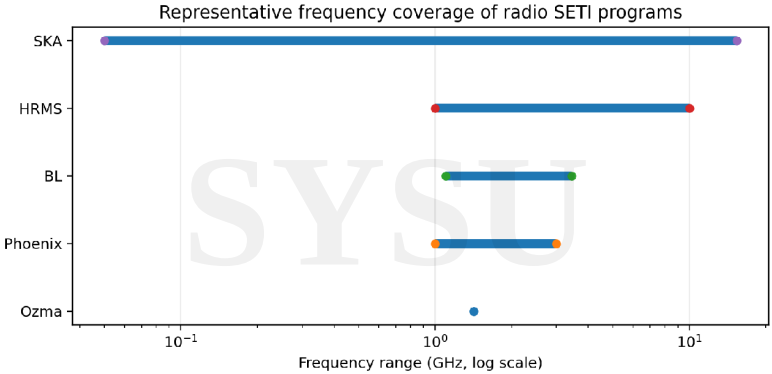

Figure 5.1: Frequency span comparison across representative radio SETI programs

Order-of-Magnitude Leap

Sky Area: From Targeted Monitoring to Cumulative Coverage

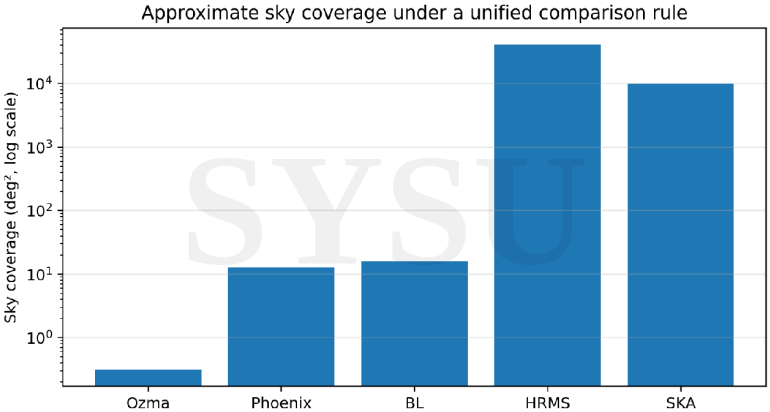

Figure 5.2: Sky coverage comparison (log scale)

Sensitivity: Beyond "Larger Aperture"

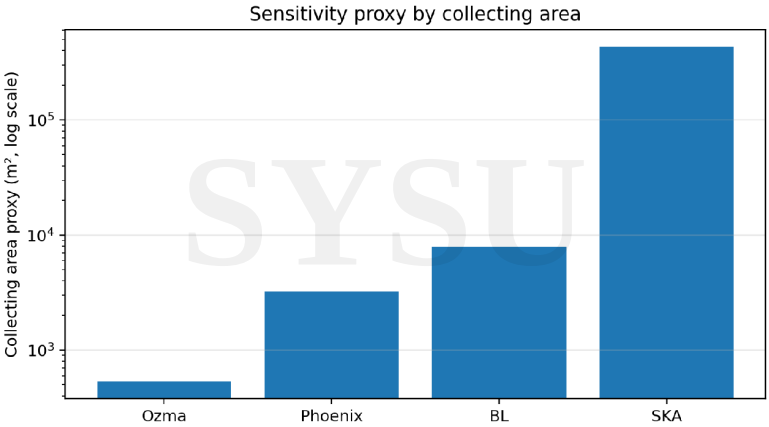

Figure 5.3: Sensitivity proxy comparison using collecting area (log scale)

Three-Dimensional Synchronous Expansion

Theorem 5.1 Core Finding

SKA may achieve **simultaneous expansion** across sky coverage, frequency coverage, and sensitivity for the first time — transforming null results from weak constraints to definitive parameter-space upper bounds.

Program	**Sky (deg^2)**	**Freq (GHz)**	**Area (m^2)**
Ozma	0.31	1.42 (single)	∼531
Phoenix	12.8	1–3	∼3,217
BL	16.0	1.10–3.45	∼7,854
NASA HRMS	4.1×10^4	1–10	multi-facility
SKA	**10^3–10^4**	**0.05–15.4**	**$\sim 4.3 \times 10^5$**

1 Introduction & Research Background

2 Methodology & Three-Dimensional Framework

3 Historical Evolution of SETI (1960–2025)

4 SETI vs. Traditional Observational Science

5 SKA's Three-Dimensional Capability Leap

6 Conclusions & Future Perspectives

7 Q & A

SYSU

Main Conclusions

1. Literature Dynamics

SETI's rise and fall are driven by **equipment capabilities, funding structures, and disciplinary context** — not by whether signals are found.

2. Three-Dimensional Framework

Sky coverage × frequency coverage × sensitivity reveals **structural differences** between historical programs and SKA.

3. SKA's Core Significance

SKA's importance lies not in guaranteeing discovery, but in **reshaping the evidence base** — pushing SETI toward parameter-space-constrained observational science.

4. Paradigm Transformation

First simultaneous expansion of three key capabilities; null results become **meaningful**

Future Perspectives

Commensal Observation

SETI backends operate in parallel with regular astronomy — no dedicated telescope time needed.

Multi-Band Coordination

Radio as primary axis; optical/IR/UV as cross-validation for technosignature searches.

Algorithm & Machine Learning

Critical for RFI rejection, candidate screening, and verification at scale.

Formalizing Null Results

Treat “no detection” as formal scientific output with standardized parameter-space reporting.

1 Introduction & Research Background

2 Methodology & Three-Dimensional Framework

3 Historical Evolution of SETI (1960–2025)

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Questions & Discussion

Thank you for your attention!

Wang Yiyu | 22344140
Supervisor: Prof. Zhang Yong
School of Physics and Astronomy, Sun Yat-sen University